

Should the Fouled Player Take the Penalty *Himself*?

A ONE-TAILED TEST · TOP 5 EUROPEAN LEAGUES

2026-05

Three sentences

THE QUESTION

When a goalkeeper fouls a player who then takes the penalty himself, does conversion drop?

THE TEST

One-tailed two-proportion z-test. Treatment $n = 100$ (75.0%). Control $n = 3,757$ (78.25%).

THE RESULT

$\Delta = -3.25$ pp (predicted direction) but $p = 0.218$, one-tailed — fail to reject H_0 at the available power.

Why we'd expect a drop

The just-fouled player walks straight to the spot.

Theory of the taker: physically jarred — body contact, hit the ground, adrenaline. Emotionally agitated, narrowed focus.

Theory of the keeper: locked in. The foul that gave the penalty was his fault. Nothing left to lose. Maximum motivation.

→ *If both pull conversion down, the bench should hand the ball to a designated specialist.*

What we built the test on

4,110

PENALTIES (LOOSE FILTER)

3,322

SINGLE-PENALTY MATCHES

100

TREATMENT-CELL N

5

LEAGUES

Seasons in scope: 2017–2018 → 2023–2024

Competitions: Premier League, La Liga, Serie A, Bundesliga, Ligue 1.

Source: FBref player–match data, consolidated locally as FULL_DICT_ROWS.pkl.

Strict filter = matches with at most one penalty event (unambiguous GK-caused / taker-was-fouled attribution).

One-tailed two-proportion z-test

TREATMENT

GK fouled the taker

Matches where the goalkeeper conceded the foul AND the fouled player is the same player who took the penalty.

CONTROL

All other penalties

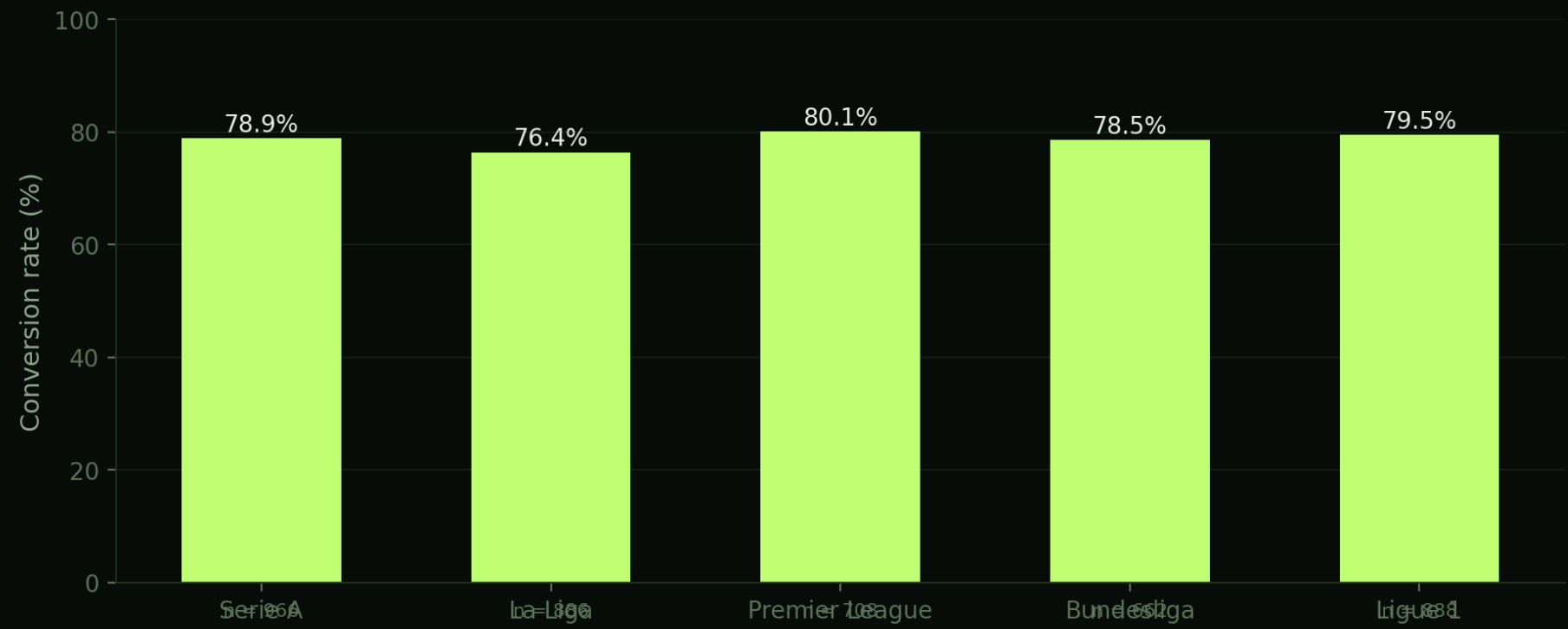
Every other strict-filter match — GK fouled but a different player took, or outfielder caused the foul (any taker).

H_0 : `conversion_treatment = conversion_control`

H_1 : `conversion_treatment < conversion_control` (one-tailed, $\alpha = 0.05$)

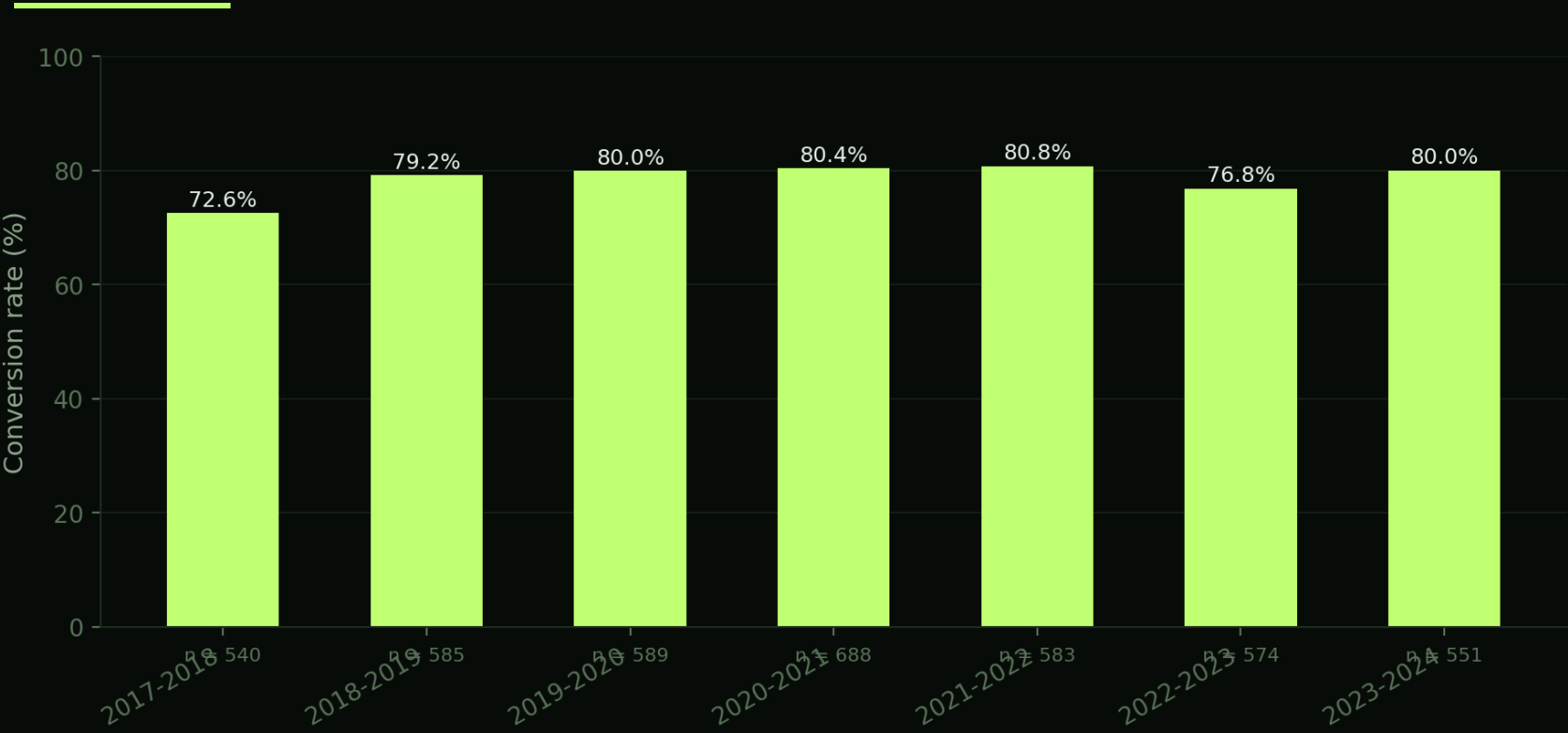
95% Wilson CIs on each proportion. Implementation: `statsmodels.proportions_ztest(alternative='smaller')`.

Conversion rate by league



Across the top 5 leagues conversion clusters tightly in the high 70s — league of origin is not a major source of variation.

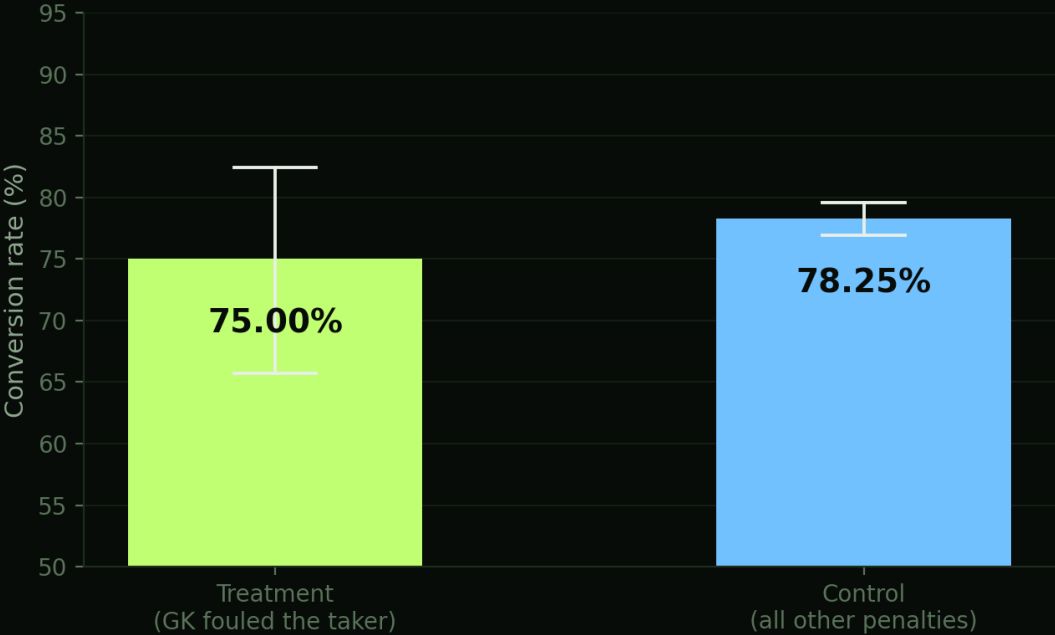
Conversion across seasons



Year-on-year stability is high — the ~77–80% band holds across the period in scope.

PRIMARY TEST · ONE-TAILED

Treatment vs control



TREATMENT

75.00%

n = 100 · 95% CI [65.70%, 82.45%]

CONTROL

78.25%

n = 3,757 · 95% CI [76.91%, 79.54%]

$\Delta = -3.25$ pp $z = -0.777$

p = 0.2185 (one-tailed) · Fail to reject H₀

Whiskers: 95% Wilson confidence intervals. Treatment CI is wide — n is the binding constraint.

2x2 CONTEXT

Where the treatment cell sits

	GOALKEEPER CAUSED THE FOUL	OUTFIELDER CAUSED THE FOUL
TAKER = THE PLAYER WHO WAS FOULED	TREATMENT 75.00% N = 100	78.87% N = 549
TAKER = A DIFFERENT PLAYER	78.11% N = 265	78.15% N = 2,943

Accent-bordered cell = treatment. The other three pool into control.

The same test, league by league

LEAGUE	T. RATE	T. N	C. RATE	C. N	Z	P (1-T)	SIG
Serie A	80.95%	21	77.70%	888	+0.354	0.6383	No · low n
La Liga	57.89%	19	77.46%	803	-2.003	0.0226	Yes · low n
Premier League	88.00%	25	78.59%	626	+1.131	0.8710	No · low n
Bundesliga	61.54%	13	78.18%	614	-1.429	0.0765	No · low n
Ligue 1	77.27%	22	79.42%	826	-0.245	0.4030	No · low n

Per-league treatment cells of 13–25 leave each test severely underpowered. Verdicts cell-by-cell are essentially noise.

Pooled test is the only inference to take seriously.

What the data says

The bench answer: *can't yet tell.*

Conversion is 3.25 pp lower when the fouled player takes the kick himself (75.00% vs 78.25% for everyone else). The point estimate moves in the predicted direction, but the treatment cell is too small to rule chance out ($p = 0.218$, one-tailed).

→ No statistical warrant to override the fouled player who wants to take it himself. Equally, no clean evidence he's at full strength.

Reproduce or dig deeper

Full methodology, source code and data:

renball.com

/PROJECTS/01-PENALTY-ANALYSIS/

- methodology.pdf · formal H_0/H_1 , test assumptions, limitations
- analysis.html · interactive dashboard, live numbers from data.json
- code/build.py · Python pipeline from raw pickles to aggregated JSON